

Beyond the Plateau

How an 8-Hour Agent Actually Works

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Not a longer chat — a different shape of work. A bigger context window won't get you there.

COMPLEX GOAL

Many steps, many files, many tools —
no single answer covers it.

NEW INFO MID-FLIGHT

Each step surfaces information the
model didn't have when it started.

ACT · VALIDATE · ADJUST

Run code, read results, revise the plan.
Repeat for hours.

WHERE IT SHOWS UP IN PRACTICE

- 1 Onboarding to an unfamiliar codebase, then continuing development
- 2 Tracing a production bug from one error message all the way to a fix
- 3 Large-scale refactor or framework migration across many files

The shared trait isn't "big task" — it's "task that needs multiple rounds, each with new context."



Long-horizon capability isn't a longer context window — it's trained behavior.

WHAT GLM-5.1 LEARNED TO DO

1

Don't plateau at 50 turns

Sustains improvement past 600 iterations · 6,000+ tool calls

2

Recover from its own errors

Async RL trains on long-horizon trajectories, not single turns

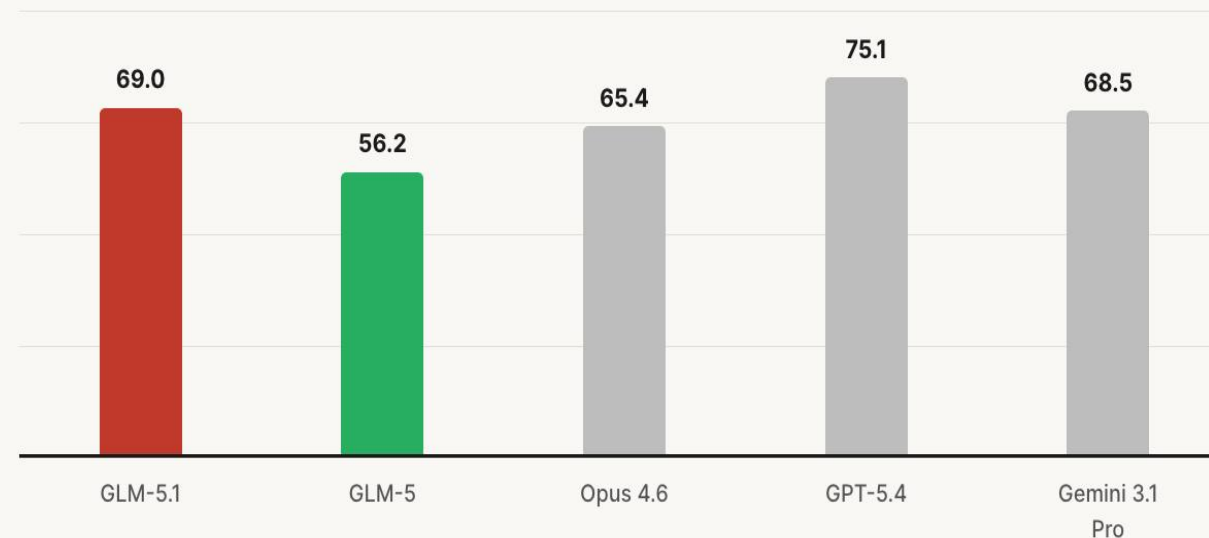
3

Self-diagnose mid-task

Reads its own logs, identifies the bottleneck, switches strategy

Real-World Terminal Tasks

Terminal Bench 2.0



Still trailing on: pure math/science reasoning · complex architecture design · GPU kernel optimization at the frontier.



THREE PRACTICES THAT MATTER MORE THAN MODEL SIZE

1 Write goals as deliverables, not visions

"Build a Linux desktop" is a wish. List modules, acceptance criteria, what counts as done.

2 Externalize state — memory belongs in the repo

goal.md · todo.md · progress.md · acceptance.md. Read every round, write every round.

3 Every round: do · verify · record

Looking busy is not the same as making progress. The most dangerous failure is drift.

THE HARNESS

Loop /loop · external loopers

State files in the repo

Tool code · shell · MCP

Eval lint · test · benchmark

Safety permissions · sandbox

A model that stays productive over hours, plus a harness that keeps the work compounding — that's the shipping form of an agent today.



Thanks!

